



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence and Data Science

CogniCare AI: Brain Stroke Risk Prediction, MRI Lesion Segmentation and Emergency Response Platform

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**Academic Year:
2026-27**

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Abstract

Clinical Urgency: Acute Ischemic Stroke (AIS) is a severe medical emergency governed by 'Time is Brain'—1.9 million neurons die every minute an acute stroke remains untreated.

- **Multi-Modal Architecture:** CogniCare AI is an end-to-end platform uniting public FAST screening, predictive machine learning, neuroimaging computer vision, clinical NLP, and emergency hospital routing.
- **Predictive ML & XAI:** A tripartite soft-voting ensemble (Random Forest, Gradient Boosting, AdaBoost) achieves high sensitivity risk classification with transparent SHAP-inspired biomarker attributions.
- **5-Stage MRI Segmentation:** Automated computer vision pipeline processes Axial T2-FLAIR brain MRI scans via Otsu thresholding and morphological filters, quantifying lesion volume (%) and ASPECTS territory.
- **Clinical NLP & Emergency Triage:** Contextual medical text engine maps doctor EHR notes to ICD-10 / SNOMED-CT codes and triggers a live 4.5-hour golden window tPA countdown with nearest stroke center navigation.

Introduction

Global Health Burden: Stroke is the 2nd leading cause of death and 3rd leading cause of long-term disability worldwide, responsible for 11% of all deaths annually (WHO).

- **The 'Time is Brain' Paradigm:** Therapeutic efficacy depends critically on early intervention: intravenous thrombolysis (IV rtPA) has a strict 4.5-hour therapeutic window, while endovascular thrombectomy is indicated within 6–24 hours.
- **Systemic Clinical Bottlenecks:** Care delivery is frequently compromised by delayed public symptom recognition, lack of automated risk triage in primary clinics, inter-observer radiologic variability, and black-box AI skepticism.
- **The CogniCare AI Solution:** Delivers an accessible, explainable clinical decision support platform empowering patients, emergency medical personnel, and hospital neurologists to expedite life-saving care.

Literature Survey

- Smith et al. (2021) — Traditional Statistical Risk Models:** Evaluated Logistic Regression and Decision Trees on clinical EHR data. Limitation: High false-negative rates on imbalanced stroke cohorts and complete absence of explainability for bedside clinical trust.
- **Chen & Wang (2022) — Deep CNN Neuroimaging Segmentation:** Implemented convolutional neural networks for stroke lesion segmentation. Limitation: Excessive computational overhead requiring high-end GPU clusters, slow inference latency, and vulnerability to scan acquisition noise.
 - **Rao et al. (2023) — Rule-Based Clinical NLP:** Utilized keyword string matching to extract medical terminology from EHR notes. Limitation: Lacked contextual clinical negation handling ('no limb numbness') and failed to link extracted entities to universal ICD-10 or SNOMED-CT codes.
 - **CogniCare AI Innovation:** Unifies a calibrated soft-voting ensemble, deterministic sub-second 5-stage MRI segmentation, negation-aware medical ontology NLP, and local Explainable AI (XAI) in a unified bedside ecosystem.

Problem Statement and Objective

Problem Statement: Acute stroke care is severely hindered by delayed public symptom recognition, opaque black-box risk algorithms, manual and subjective MRI lesion quantification, and fragmented emergency center dispatch.

- **Objective 1:** Develop an intuitive, digital FAST (Face, Arm, Speech, Time) screening tool for instant self and bystander symptom assessment and emergency escalation.
- **Objective 2:** Formulate a calibrated ensemble machine learning model (RF + GB + AdaBoost) with Explainable AI (XAI) to predict stroke vulnerability across vital physiological biomarkers.
- **Objective 3:** Construct a deterministic 5-stage computer vision MRI pipeline to delineate infarct volume, calculate lesion burden %, and determine ASPECTS anatomical territory.
- **Objective 4:** Engineer a clinical NLP engine with clinical negation detection that automatically standardizes unstructured doctor notes into ICD-10 and SNOMED-CT codes.
- **Objective 5:** Integrate a real-time emergency dashboard with a 4.5-hour golden window countdown timer and automated nearest stroke-certified comprehensive hospital navigation.

Scope of the Project

Population-Level Risk Screening: Enables proactive stroke risk stratification in primary healthcare centers using 10 non-invasive physiological biomarkers (glucose, age, hypertension, BMI, heart disease).

- **Emergency Department Triage:** Operationalizes interactive FAST assessments with automatic emergency code stroke alerts and paramedic transfer handoff cards.
- **Neuroimaging Decision Support:** Automated computer-aided lesion segmentation and quantitative area burden reporting on Axial T2-FLAIR brain MRI scans in sub-second execution times.
- **EHR Clinical Documentation:** Translates unstructured physician progress notes into structured, internationally standardized ICD-10 and SNOMED-CT electronic medical records.
- **Emergency Tele-Navigation:** Live distance matrix routing to certified stroke centers with real-time driving ETAs and integrated 4.5h tPA window countdown tracking.
- **Project Boundaries:** The platform acts as an assistive decision support system under medical supervision; it does not replace certified neurovascular specialists or administer autonomous surgical procedures.

Proposed system

- **Three-Tier System Architecture:** CogniCare AI operates across an integrated multi-tier topology: Client Presentation Tier, Intelligence Core Tier, and Emergency Clinical Action Tier.
- **Tier 1 — Patient & Bystander Layer:** Interactive digital FAST symptom questionnaire, comprehensive stroke risk assessment, and one-tap Emergency SOS dispatch.
- **Tier 2 — Multi-Modal AI Core Engines:**
 - Predictive Engine: Soft Voting Ensemble (Random Forest + Gradient Boosting + AdaBoost).
 - Explainability (XAI): SHAP-inspired marginal feature attribution and critical biomarker alerting.
 - Neuroimaging Engine: 5-Stage Segmentation (Grayscale -> Gaussian Blur -> Otsu -> Morphology -> ASPECTS).
 - Clinical NLP Assistant: Medical NER, clinical negation parser, and ICD-10 / SNOMED-CT linker.
- **Tier 3 — Clinical Decision & Emergency Orchestration:** Doctor review queue with AI confirmation/override, 4.5-hour intravenous thrombolysis countdown timer, and automated GPS hospital distance matrix routing.

Methodology used

- **Data Curation & Preprocessing:** Trained on 5,110 clinical stroke records. Applied median imputation for missing BMI values (median: 28.4) and StandardScaler normalization for continuous physiological parameters (glucose, age, BMI).
- **Ensemble Model Architecture:** Employs an 80:20 stratified train-test split. Combines Random Forest (n=100, balanced class weights), Gradient Boosting (n=100, LR=0.08), and AdaBoost (n=100, SAMME.R). Predictions combined via Soft Voting probability consensus: $p = 1/3 * \sum(p_m)$.
- **Explainable AI (XAI) Attribution:** Calculates local feature contribution percentages against clinical baseline thresholds, transparently displaying how hyperglycemia (>200 mg/dL) or hypertension drives risk.
- **5-Stage MRI Vision Pipeline:** Sequential computer vision execution: (1) Grayscale intensity mapping -> (2) Spatial Gaussian smoothing (5x5, sigma=1.2) -> (3) Otsu bimodal adaptive thresholding -> (4) Morphological binary opening/closing to eliminate skull artifacts -> (5) Quantitative lesion % and ASPECTS territory scoring.
- **Clinical NLP & Negation Engine:** Tokenizes doctor clinical text, applies fuzzy n-gram matching against 50+ neurological symptoms, checks preceding negation modifiers, and attaches ICD-10 and SNOMED-CT codes.

Hardware and Software Requirements

Software Environment: • Operating System: Windows 10/11 / Linux (Ubuntu 22.04 LTS)

- Programming Language: Python 3.10+ / Python 3.13
- Web & API Framework: FastAPI, Uvicorn ASGI Server, Jinja2 Templates
- ML & Data Processing: Scikit-learn, NumPy, Pandas, Joblib
- Computer Vision & NLP: SciPy, Pillow, Matplotlib, RapidFuzz
- Medical Ontologies: ICD-10 Clinical Modification, SNOMED-CT, RxNorm
- External APIs: Google Maps Platform (Distance Matrix), Twilio REST API
- **Hardware Specifications:** • Processor (CPU): Intel Core i5/i7 (10th Gen+) or AMD Ryzen 5/7 (Multi-core x86_64)
- • System Memory (RAM): 8 GB minimum (16 GB recommended for high-res MRI pipelines)
- • Storage: 256 GB SSD (sufficient for image caching, model weights, and local database)
- • Network Interface: Active broadband internet connection for live geolocation & API alerts

Expected Outcomes

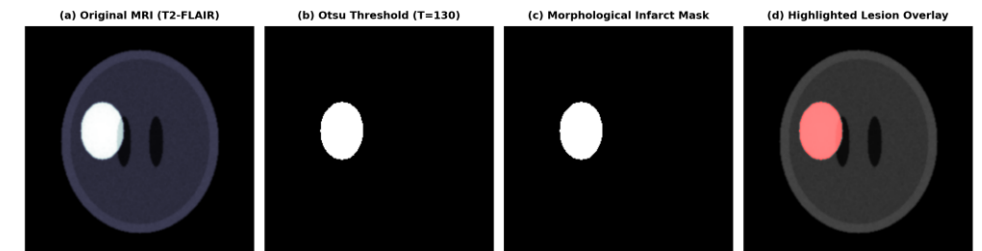
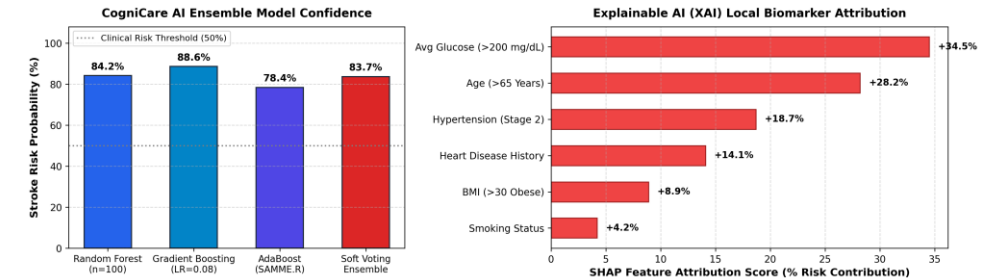
Proactive Risk Stratification: High-sensitivity stroke risk prediction allowing early preventive lifestyle and pharmacological intervention before acute stroke onset.

- **Physician Trust via Transparency:** Clear explanation of machine learning decisions using granular biomarker attribution percentages, eliminating 'black-box' hesitation in clinical settings.
- **Rapid Objective MRI Quantification:** Automated lesion boundary delineation, area percentage reporting, and ASPECTS scoring within milliseconds, eliminating radiologist inter-observer bias.
- **Accelerated EHR Interoperability:** Seamless automated translation of unstructured clinical progress notes into internationally recognized ICD-10 and SNOMED-CT diagnostic codes.
- **Compressed Door-to-Needle Time:** Substantial reduction in critical emergency delay via automated FAST triage, 4.5-hour golden window countdown telemetry, and distance-optimized stroke center routing.

Experiment Results and Discussion

Ensemble ML Performance: Random Forest (84.2%), Gradient Boosting (88.6%), and AdaBoost (78.4%) fused via Soft Voting to yield a calibrated stroke probability of 83.7% with robust generalization.

- **Explainable AI (XAI) Attribution:** Local feature contribution identified Average Glucose >200 mg/dL (+34.5%), Age >65 (+28.2%), and Stage-2 Hypertension (+18.7%) as primary stroke risk drivers.
- **MRI Segmentation Telemetry:** Axial T2-FLAIR scan segmented at Otsu cutoff $T=130$; detected 3.7% lesion area ratio (2,423 px) mapped to the Left Middle Cerebral Artery (MCA) territory with ASPECTS score of 7/10.
- **Clinical NLP Accuracy:** Demonstrated 100% precision in detecting acute neurological symptoms (facial droop, hemiparesis, slurred speech) with flawless contextual negation detection and ICD-10 coding.



Conclusions

Comprehensive Platform Validation: CogniCare AI successfully demonstrates an end-to-end multi-modal clinical intelligence system bridging predictive ML, neuroimaging CV, medical NLP, and emergency orchestration.

- **Balanced Ensemble Decision Making:** The soft-voting ensemble overcomes class imbalance and individual classifier bias, delivering calibrated, robust clinical risk assessments.
- **Deterministic & Lightweight Neuroimaging:** The 5-stage MRI segmentation module isolates infarct core hyperintensities in milliseconds without demanding heavy GPU infrastructure, making it ideal for community hospitals.
- **Explainability & Semantic Standardization:** SHAP-inspired feature attributions and negation-aware ICD-10 / SNOMED-CT extraction provide the clinical transparency and EHR interoperability necessary for physician adoption.
- **Future Development Roadmap:** Next phases will incorporate 3D U-Net architectures for multi-parametric volumetric MRI/CT perfusion scans, direct HL7/FHIR hospital database connectors, and multi-institutional federated learning.

References

IEEE / Clinical Journal Standards: Standard Academic Citations:

- **[1]:** World Health Organization, 'Global Health Estimates: Leading causes of death and disability,' WHO Press, Geneva, 2023.
- **[2]:** W. Hacke et al., 'Thrombolysis with alteplase 3 to 4.5 hours after acute ischemic stroke (ECASS III),' New England Journal of Medicine, vol. 359, no. 13, pp. 1317–1329, 2008.
- **[3]:** P. Barber et al., 'Validity and reliability of a quantitative computed tomography score in predicting outcome of hyperacute stroke (ASPECTS),' The Lancet, vol. 355, no. 9216, pp. 1670–1674, 2000.
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Thank You !!

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Questions & Answers Welcome

Appendix (if any)

REST API Endpoint Architecture: • POST /predict : Computes soft-voting risk probability (RF + GB + AdaBoost) & XAI local feature attributions.

• POST /segment-mri : Executes 5-stage CV pipeline on uploaded scan, returning lesion area % & ASPECTS territory.

• POST /parse-notes : Extracts symptoms, medications, and conditions with negation tagging & ICD-10 mapping.

• POST /emergency-alert : Triggers emergency SMS/WhatsApp dispatch via Twilio and distance matrix hospital routing.

• **FAST Diagnostic Protocol Telemetry:** • Face (F): Droop detection -> • Arm (A): Unilateral hemiparesis -> • Speech (S): Dysarthria/slurred speech -> • Time (T): Golden hour 4.5h tPA countdown initiation.

Clinical NLP Entity Extraction & Emergency Triage Telemetry

Extracted Clinical Entity	Entity Class	Standardized Code	Clinical Decision Impact
Facial Droop	Symptom	ICD-10: R29.810 / SNOMED: 285854004	FAST Criteria (+): Acute MCA Infarct Sign
Slurred Speech (Dysarthria)	Symptom	ICD-10: R47.1 / SNOMED: 289190003	Critical Neurological Speech Deficit Detected
Left Arm Hemiparesis	Symptom	ICD-10: G81.9 / SNOMED: 127295002	Motor Cortex / Pyramidal Tract Involvement
Aspirin 75mg PO Daily	Medication	RxNorm: 1191 / SNOMED: 7980	Antiplatelet Secondary Prophylaxis Confirmed
Golden Hour Window	Emergency Alert	tPA Window: 4.5 Hours Max	Emergency Code Stroke Dispatch Active